

AMRUTH SHYJU

Backend-Focused MERN Stack Developer

Node.js • TypeScript • Express.js • MongoDB • Redis • Scalable Backend Systems

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Professional Summary

Backend Developer specializing in Node.js and TypeScript with hands-on experience building scalable backend applications using MongoDB, Redis, BullMQ, and Socket.IO. Skilled in designing modular systems using Clean Architecture, developing REST APIs, implementing real-time communication, and deploying applications on AWS.

Technical Skills

Languages: JavaScript (ES6+), TypeScript

Backend: Node.js, Express.js, REST APIs, JWT Authentication, RBAC

Databases: MongoDB (Aggregation, Transactions, Indexing, Data Modeling), SQL

Real-Time & Messaging: Socket.IO, WebSockets, Redis, BullMQ, Pub/Sub

Architecture: Clean Architecture, Repository Pattern, Dependency Injection, Event-Driven Architecture

Frontend: React.js, Next.js, TanStack Query, Leaflet

DevOps & Cloud: AWS EC2, Docker, Nginx, GitHub Actions, CI/CD

Security: Zod Validation, Helmet, Express Rate Limit, HTTP-only Cookies

Tools: Git, GitHub, Postman, Jira, Figma

Projects

SPEEDO – GPS Fleet Trip Analytics System

[Live](#) | [GitHub](#)

Node.js, TypeScript, MongoDB, Redis, BullMQ, Socket.IO, React, Leaflet

- Built a real-time GPS telemetry pipeline using **BullMQ, Redis, and Socket.IO** to process vehicle location updates asynchronously with retry mechanisms and live trip synchronization.
- Developed a modular backend using **TypeScript, Clean Architecture, dependency injection, and MongoDB transactions**, improving maintainability and ensuring reliable trip-processing workflows.
- Implemented geospatial analytics features including **trip tracking, overspeed detection, stoppage analysis, and driver behavior insights** using optimized **O(n)** algorithms and Haversine distance calculations for efficient route analysis.
- Built an interactive **React 19 + Leaflet** dashboard to visualize live vehicle locations, trip playback, and thousands of GPS points with real-time updates using Socket.IO and TanStack Query.

STRATIFY – Multi-Tenant Workforce & Project Management SaaS

[Live](#) | [GitHub](#)

Node.js, TypeScript, React, MongoDB, Redis, Socket.IO, AWS

- Built a multi-tenant SaaS application using Node.js and Clean Architecture, isolating company data with **AsyncLocalStorage** and a custom **Mongoose plugin** that automatically applies tenant filters to database queries.
- Implemented secure authentication and authorization using **JWT**, hierarchical **RBAC**, centralized request validation, rate limiting, and Helmet to protect APIs and enforce tenant-level access control.
- Developed real-time collaboration features including one-to-one chat, group messaging, notifications, and video meetings using **Socket.IO**, Redis Pub/Sub, and ZegoCloud.
- Built project and workforce management modules including Kanban boards, leave management, task assignment, and Razorpay subscription integration with Cloudinary-based media uploads.

Education

Professional Training– Full Stack Development (MERN Stack)- Brototype

2024 – Present

- Built full-stack applications using **Node.js, Express.js, React, TypeScript, and MongoDB**, focusing on scalable backend development and REST API design.
- Implemented backend features including authentication, role-based access control, real-time communication with Socket.IO, background job processing, and third-party service integrations.
- Practiced software engineering principles including Clean Architecture, Dependency Injection, Git-based collaboration, debugging, deployment on AWS EC2, and code optimization through hands-on projects.